# Generated benchmark artifacts

Typeset from paper/generated/aarch64/ by benchmarks/scripts/build_pdf.py. Every number resolves to a row in the result set named below; see PROVENANCE.md for the inputs, transformation and caveats of each artifact.

result set: current
commit: 9b9dcd5
host: galaxy
measured: 2026-09-07
inputs: sha256:672bb55f5e5a7c9f

| Dataset | Tool | Metric | -@ | Mapped | Mapq 60 | Missed (%) | Wrong Q60 | Index (s) | Map (s) | Mem (GB) |
|---|---|---|---|---|---|---|---|---|---|---|
| B01 | shmap-rs | Containment | 1 | 149 194 | 139 309 | 6.78 | — | 6.52 | 36.27 | 2.66 |
| B01 | shmap-rs | Jaccard | 1 | 146 120 | 138 421 | 7.37 | — | 6.50 | 50.77 | 2.65 |
| B01 | shmap-rs | bucket_SH | 1 | 149 236 | 138 250 | 7.49 | — | 6.53 | 27.77 | 2.63 |
| B01 | cpp-shmap | Containment | 1 | 149 194 | 139 309 | 6.78 | — | 25.42 | 77.67 | 18.85 |
| B01 | cpp-shmap | Jaccard | 1 | 146 120 | 138 421 | 7.37 | — | 25.34 | 105.03 | 18.85 |
| B01 | cpp-shmap | bucket_SH | 1 | 149 236 | 138 250 | 7.49 | — | 25.64 | 61.44 | 18.85 |
| B01 | winnowmap2 | — | 32 | 189 022 | — | — | — | — | 1493.40 | 9.83 |
| B02 | shmap-rs | Containment | 1 | 125 000 | 117 532 | 5.97 | 0 | 6.54 | 27.82 | 2.66 |
| B02 | shmap-rs | Jaccard | 1 | 125 000 | 119 065 | 4.75 | 6 | 6.50 | 43.53 | 2.65 |
| B02 | shmap-rs | bucket_SH | 1 | 125 000 | 116 800 | 6.56 | 1 | 6.54 | 23.97 | 2.66 |
| B02 | cpp-shmap | Containment | 1 | 125 000 | 117 532 | 5.97 | — | 25.60 | 60.76 | 18.85 |
| B02 | cpp-shmap | Jaccard | 1 | 125 000 | 119 065 | 4.75 | — | 25.08 | 90.63 | 18.85 |
| B02 | cpp-shmap | bucket_SH | 1 | 125 000 | 116 802 | 6.56 | — | 25.12 | 54.63 | 18.85 |
| B02 | winnowmap2 | — | 32 | 129 488 | — | — | — | — | 1123.80 | 9.47 |
| B03 | shmap-rs | Containment | 1 | 241 991 | 227 521 | 6.19 | — | 6.49 | 45.23 | 2.65 |
| B03 | shmap-rs | Jaccard | 1 | 241 265 | 228 887 | 5.63 | — | 6.53 | 61.01 | 2.65 |
| B03 | shmap-rs | bucket_SH | 1 | 242 050 | 226 931 | 6.43 | — | 6.51 | 36.23 | 2.65 |
| B03 | cpp-shmap | Containment | 1 | 241 991 | 227 521 | 6.19 | — | 25.57 | 86.53 | 18.85 |
| B03 | cpp-shmap | Jaccard | 1 | 241 265 | 228 886 | 5.63 | — | 25.68 | 117.10 | 18.85 |
| B03 | cpp-shmap | bucket_SH | 1 | 242 050 | 226 934 | 6.43 | — | 26.06 | 69.73 | 18.85 |
| B03 | winnowmap2 | — | 32 | 280 093 | — | — | — | — | 1113.10 | 8.22 |
| B04 | shmap-rs | Containment | 1 | 2 419 796 | 2 273 122 | 6.28 | — | 6.54 | 455.45 | 2.66 |
| B04 | shmap-rs | Jaccard | 1 | 2 411 461 | 2 286 085 | 5.74 | — | 6.52 | 622.92 | 2.65 |
| B04 | shmap-rs | bucket_SH | 1 | 2 420 331 | 2 267 229 | 6.52 | — | 6.51 | 360.70 | 2.65 |
| B04 | cpp-shmap | Containment | 1 | 2 419 796 | 2 273 121 | 6.28 | — | 25.04 | 877.05 | 18.85 |
| B04 | cpp-shmap | Jaccard | 1 | 2 411 461 | 2 286 083 | 5.74 | — | 25.12 | 1195.37 | 18.85 |
| B04 | cpp-shmap | bucket_SH | 1 | 2 420 331 | 2 267 238 | 6.52 | — | 25.12 | 703.13 | 18.85 |
| B04 | winnowmap2 | — | 32 | 2 808 979 | — | — | — | — | 10434.70 | 8.47 |
| B05 | shmap-rs | Containment | 1 | 39 608 | 37 839 | 58.97 | — | 6.51 | 13.68 | 2.65 |
| B05 | shmap-rs | Jaccard | 1 | 5 956 | 5 409 | 94.13 | — | 6.54 | 16.80 | 2.66 |
| B05 | shmap-rs | bucket_SH | 1 | 41 115 | 39 226 | 57.46 | — | 6.49 | 12.57 | 2.65 |
| B05 | cpp-shmap | Containment | 1 | 39 608 | 37 839 | 58.97 | — | 25.18 | 26.58 | 18.85 |
| B05 | cpp-shmap | Jaccard | 1 | 5 956 | 5 409 | 94.13 | — | 25.15 | 31.18 | 18.85 |
| B05 | cpp-shmap | bucket_SH | 1 | 41 115 | 39 225 | 57.47 | — | 25.21 | 25.02 | 18.84 |
| B05 | winnowmap2 | — | 32 | 99 107 | — | — | — | — | 327.10 | 6.11 |

Table 1: Long-read mapping tools on the benchmark suite. Mapq 60 is the count of confident mappings; Missed is the share of input reads either unmapped or below mapq 60; Wrong Q60 counts confident mappings that are wrong against ground truth.

| Dataset | Metric | Matches per read | | | Potential | | Buckets per read | |
|---|---|---|---|---|---|---|---|---|
| | | possible | examined | in mapping | realized | unrealized | seeded | final |

Table 2: Efficiency of the seed heuristic. Realized potential is possible matches divided by matches examined (work avoided); unrealized potential is matches examined divided by matches inside the reported mapping (work still wasted).

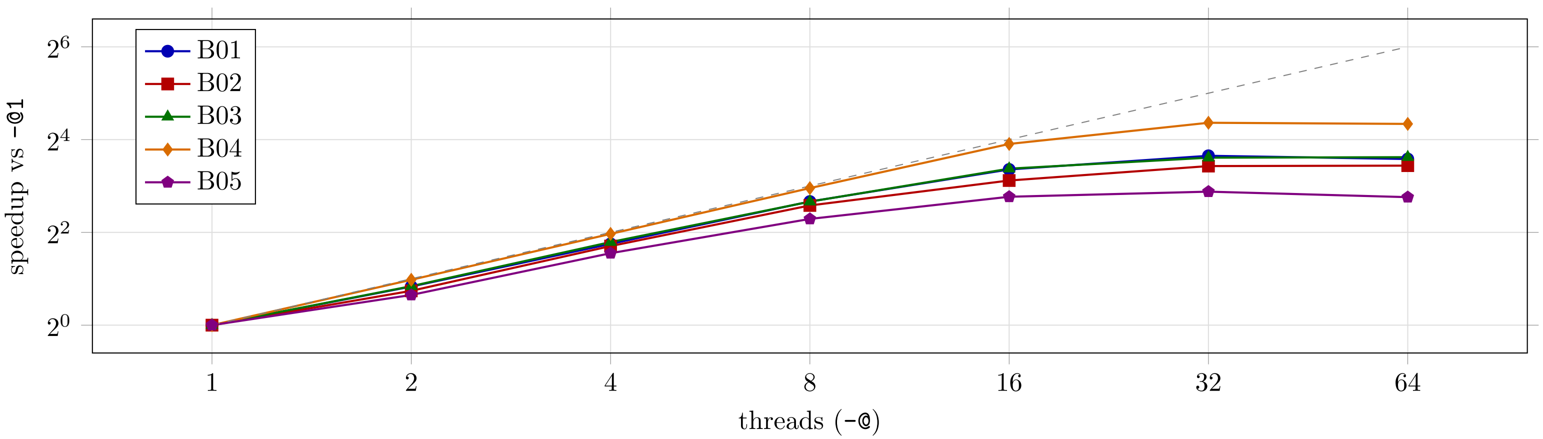


Figure 1: Whole-run speedup against thread count, Containment. The dashed line is linear speedup.

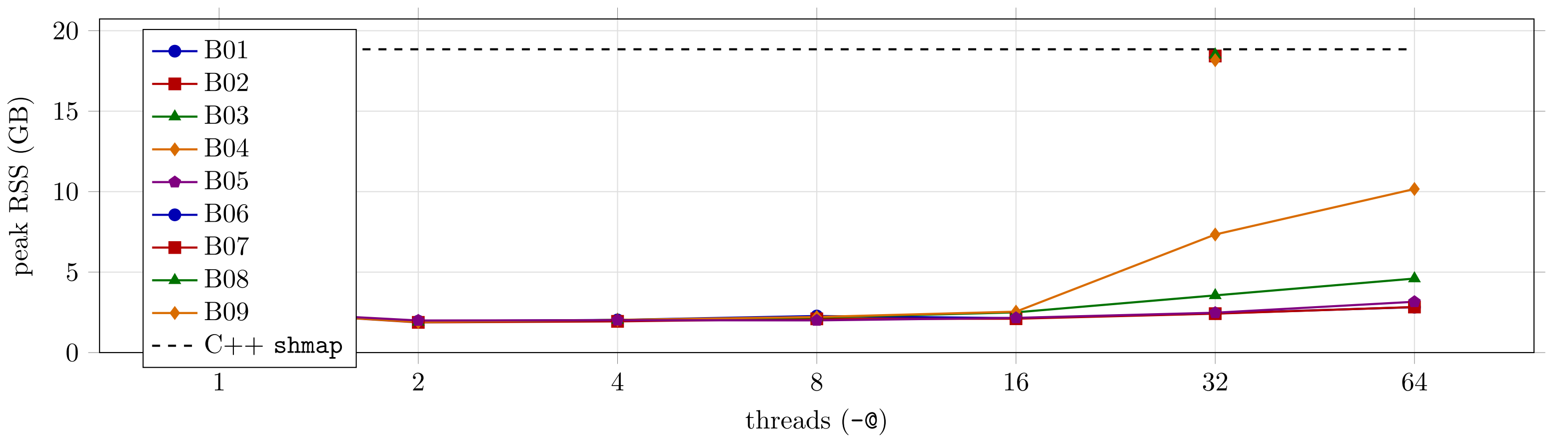


Figure 2: Peak resident memory against thread count, Containment, with the C++ reference as a horizontal line.

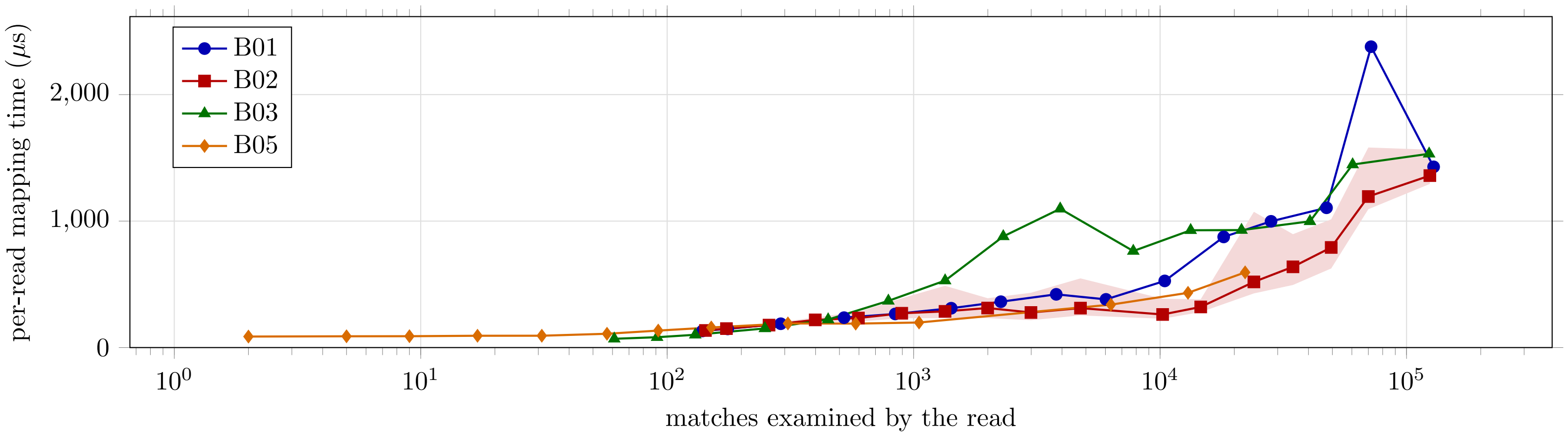


Figure 3: Per-read mapping time against the number of matches the read examined, Containment. Points are per-bin medians over reads grouped into 18 logarithmic bins; the shaded band is the interquartile range for the simulated set.

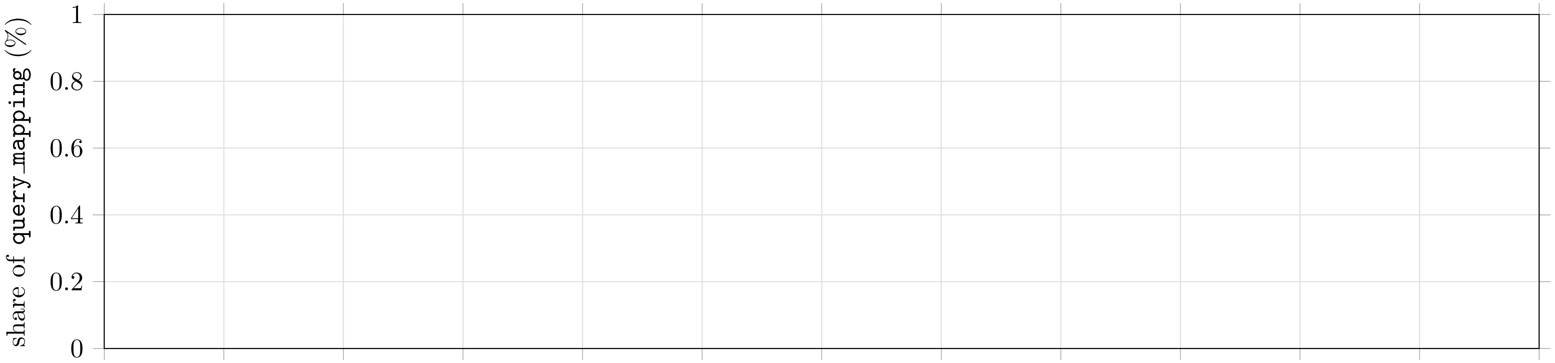


Figure 4: Where mapping time goes: stage shares of `query_mapping`, single-threaded.

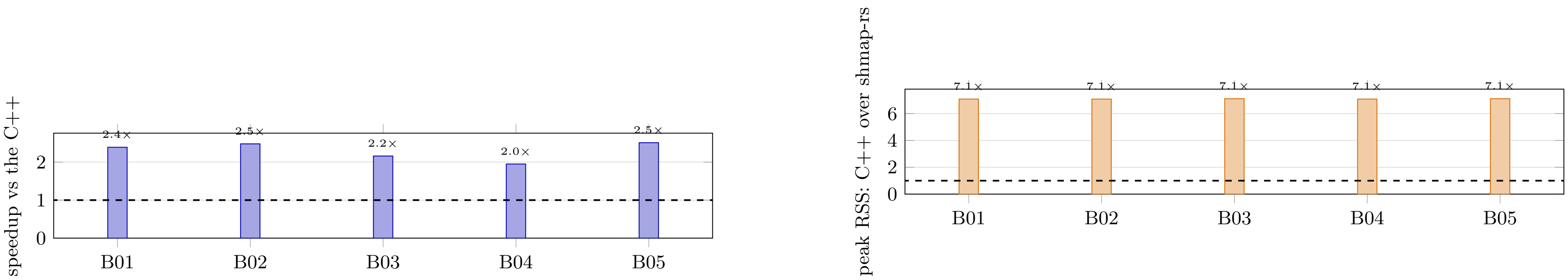


Figure 5: The two headline results, per benchmark, single-threaded on both sides: wall-clock speedup over the C++ (top) and how many times less peak memory the port uses (bottom). The dashed line is parity.

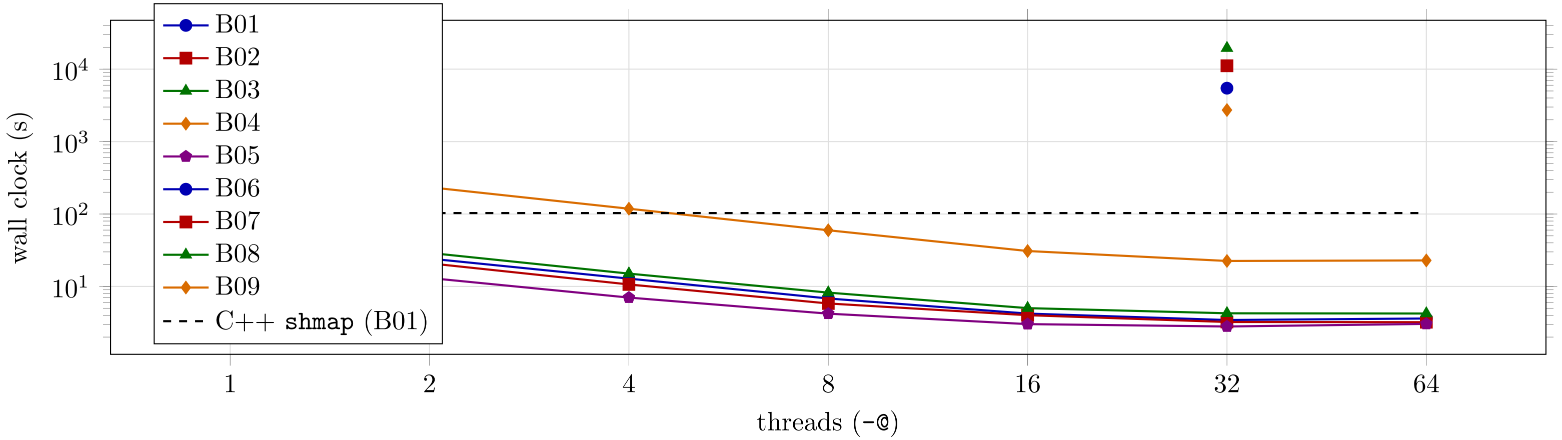


Figure 6: Wall clock against thread count, Containment, with the C++ reference as a horizontal line. The gap at `-@1` is the single-threaded speedup the summary quotes; everything to the right of it is threading the C++ does not have.